AUTOPILOTS ON AZURE / ARCHITECTURE

Technical architecture

Infrastructure, application and logical diagrams

01 / INFRASTRUCTURE

Where compute runs and traffic crosses boundaries.

02 / APPLICATION

Which component reasons, stores state, and calls tools.

03 / CONVERSATION

What happens after an explicit Teams mention.

04 / DOCUMENTS

Where edits happen and how publication survives a lock.

READING THE MAP

The gateway connects; the Worker does the work.

A **Worker** is a named digital colleague with its own identity, role, private state, and tools. Its **gateway** accepts authenticated activities and manages runtime availability. The **agent runtime** owns reasoning, tool use, memory, skills, and document work. Foundry supplies model inference; it does not host this agent's compute.

Solid: design connection Dashed: optional connection or component

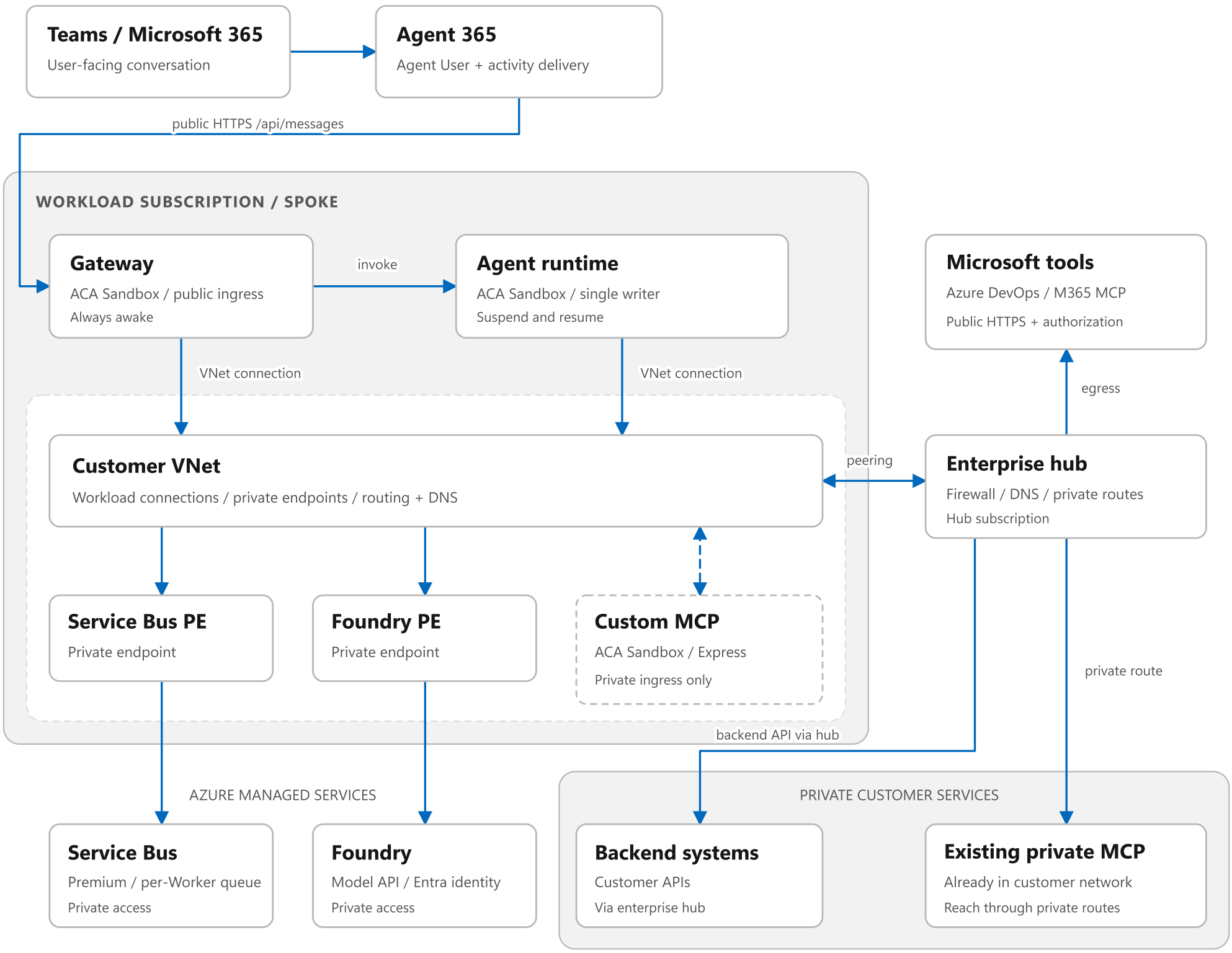
01

Infrastructure diagram

The workload VNet connects compute, private services, and the enterprise hub.

Agent 365 reaches the gateway at public HTTPS `POST /api/messages`. Gateway and agent runtime connect to the workload VNet. Service Bus and Foundry use private endpoints. The VNet connects to the enterprise hub for internal services and controlled internet egress.

01 / Infrastructure and network boundaries



PE means private endpoint. Custom MCP reaches backend systems through the workload VNet, peering, and enterprise hub before entering the private customer network.

Tools fit three patterns: Microsoft-hosted MCP services on public HTTPS, existing MCP services in the customer network, or Custom MCP deployed with the solution. Custom MCP exposes a private interface and calls backend APIs through the enterprise hub. Choose a supported private-ingress ACA configuration.

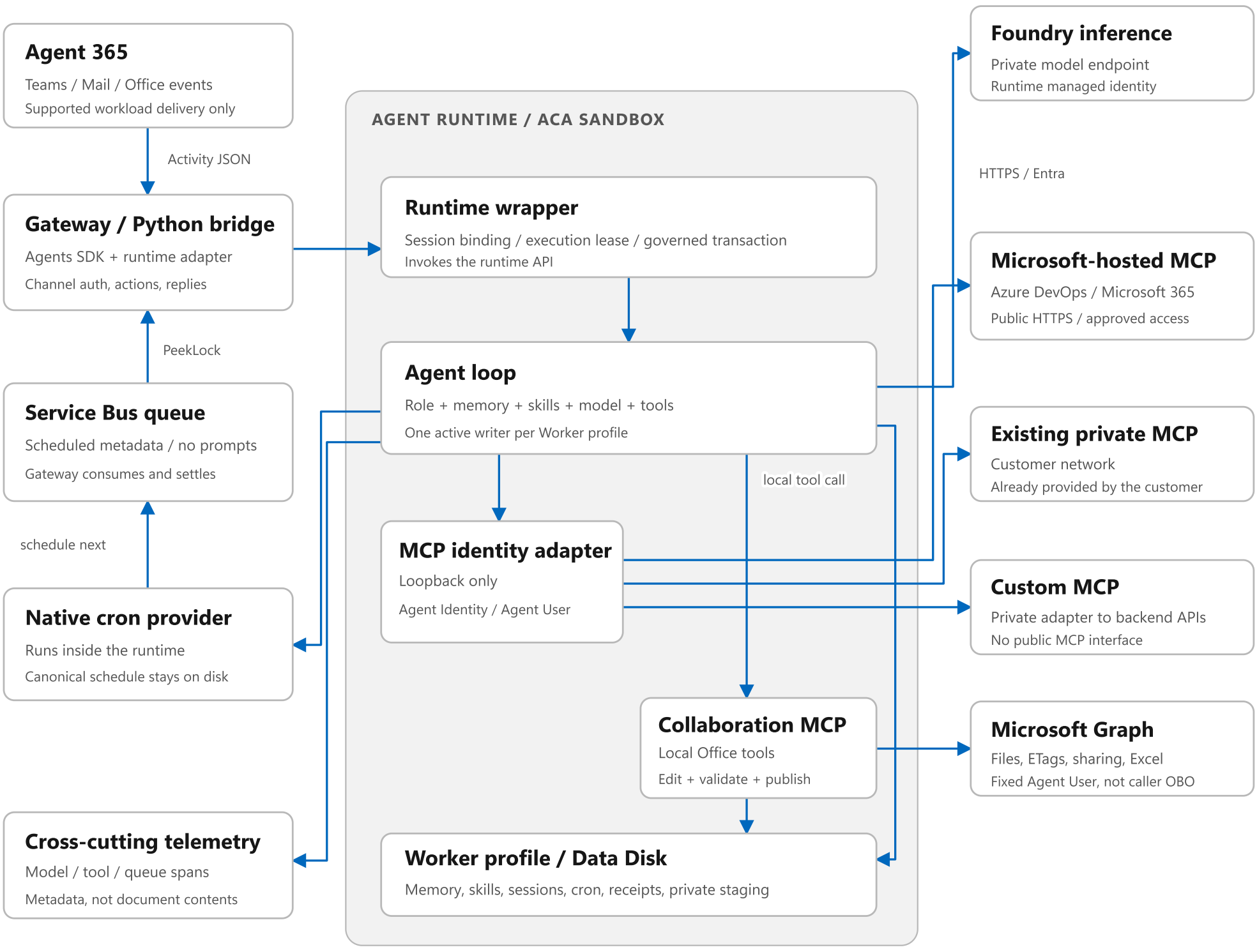
02

Application architecture

Model calls, MCP, Office editing, and persistent state meet inside the runtime.

The gateway normalizes authenticated channel activities, binds the conversation, validates interactive actions, and wakes or reuses the runtime. A runtime wrapper adds execution and learning boundaries around the agent loop. Model and tool traffic leaves the runtime directly; the gateway is neither an inference proxy nor a general tool router.

02 / Application responsibilities and interfaces



The cron box is drawn outside for clarity; its code executes in the runtime.

Arrows show calls or trigger delivery. Tool access follows the three integration patterns in the infrastructure view; telemetry spans the stack.

03

The queue contains wake-up metadata, not the agent's private context.

Three message types, one awake consumer, explicit settlement.

The agent runtime keeps schedules and task content on the Data Disk. Its Azure cron provider schedules the next occurrence in Service Bus. The gateway continuously receives messages under PeekLock, renews the lock, checks the target Worker, invokes the appropriate runtime operation, and explicitly completes, abandons, or dead-letters the message.

MESSAGE TYPE	PRODUCER / CREDENTIAL	CONSUMER ACTION
`hermes.cron.fire`	Runtime cron provider / federated Agent Identity	Claim the due occurrence, run the saved task, record output, deliver to its bound conversation, and reconcile the next occurrence.
`system.dream`	Runtime cron provider / federated Agent Identity	Run the reserved reflection workflow with fenced checkpoints. Ambiguous already-started work stops for inspection rather than blindly running again.
`document.publish.retry`	Gateway / gateway managed identity	Resume a saved publication operation through a fixed API. No new model turn is needed.

04

A network route never substitutes for an identity decision.

Keep infrastructure, autonomous agent, and Microsoft 365 user permissions distinct.

CALL	PRINCIPAL	ENFORCEMENT
Gateway to runtime lifecycle	Gateway managed identity	Sandbox Group data-plane role on the intended runtime Group.
Agent runtime to Foundry	Runtime managed identity	Entra token and model-resource RBAC, through the native provider.
Agent runtime to autonomous custom MCP	Federated Agent Identity	Expected issuer, tenant, audience, Worker binding, and application role.
Work IQ / Graph	Fixed Agent User via Agent Identity exchange	Approved delegated permissions and resource sharing. This is not human on-behalf-of access.
Service Bus receive / document retry send	Gateway managed identity	Queue-scoped Data Receiver / Data Sender roles.

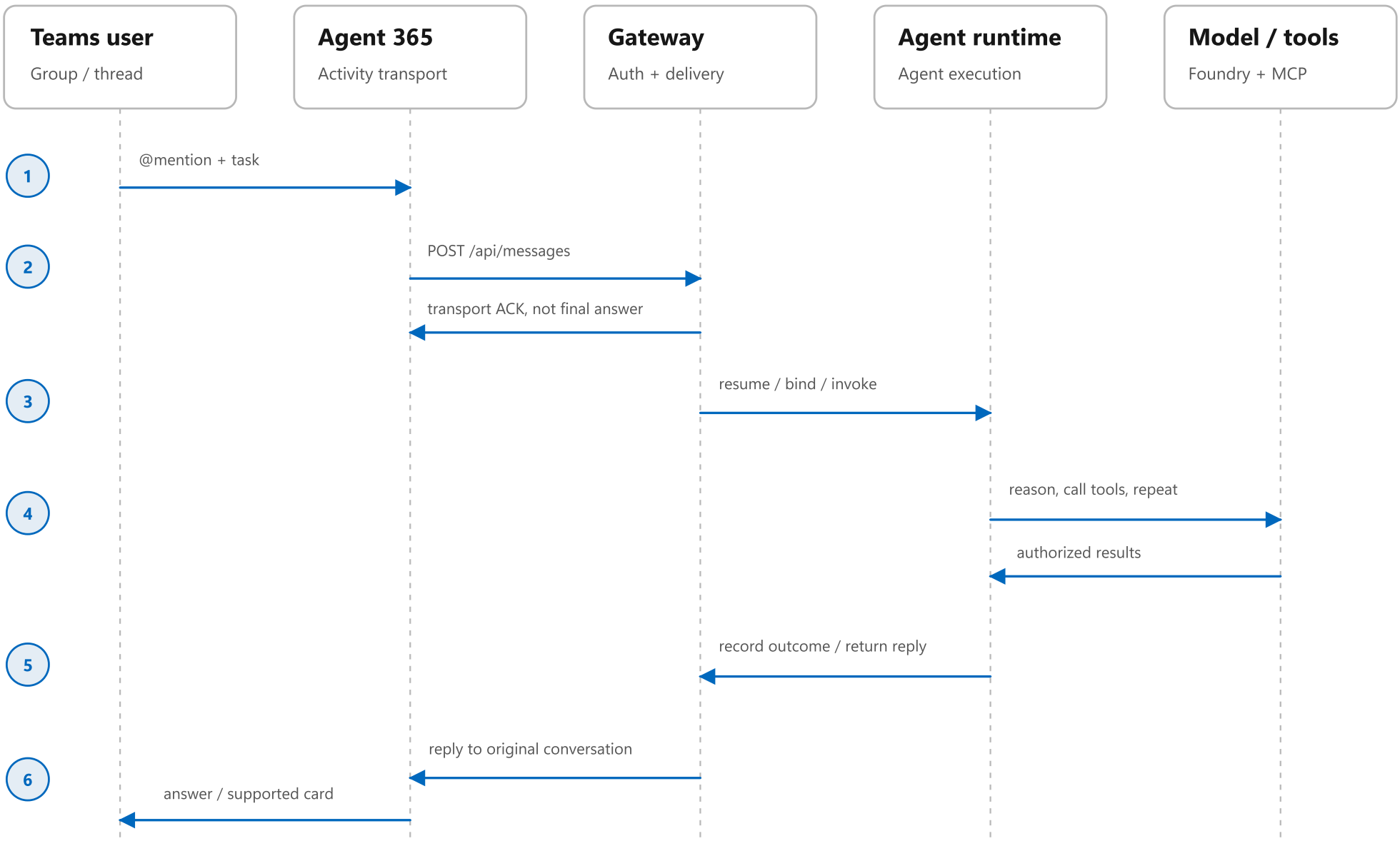
05

Teams message lifecycle

Illustrative request: “@Project Worker, summarize the delivery risks.”

The user explicitly mentions the Agent User in a group conversation. This example assumes that the host delivers the activity to the registered Worker endpoint. The gateway does not subscribe to all channel traffic and cannot assume complete group history or delivery of every event supported by the SDK.

03 / Interactive Teams request



Service Bus enters only when the task creates scheduled work or a publication retry.

The order illustrates one supported interaction, not an exact packet capture or latency promise. The model/tool loop can take several calls. Public group output must respect the conversation's audience.

1 **Authenticate before invoking the Worker**

The Agents SDK validates Activity Protocol authentication. The bridge resolves the Worker, caller, and conversation; unsupported or invalid input does not become an agent task.

2 **Reuse state without allowing concurrent writers**

The gateway ensures the runtime is available. The wrapper binds native session context and obtains the execution boundary; the agent runtime reads the appropriate profile and performs the requested work.

3 **Reply through the originating workload**

The bridge delivers the result through the channel. Office comment requests stay in their comment thread. An early HTTP acknowledgement is not a durable queue entry or proof of task completion.

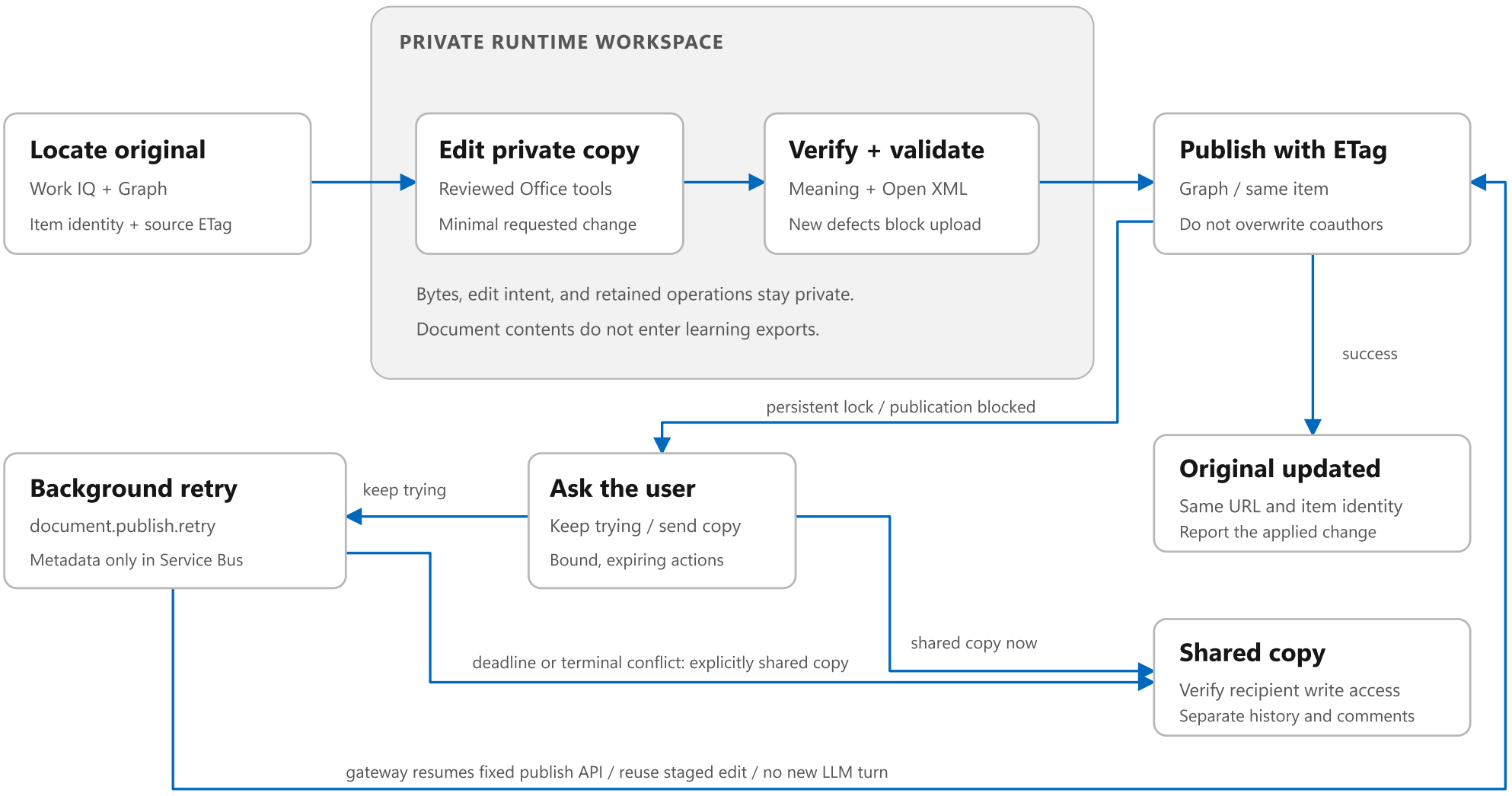
06

Document editing

Illustrative request: update a shared Word status report and explain the changes.

Work IQ discovers the shared document and supports semantic reads, comments, and other available operations. For an existing Word body edit, reviewed tooling inside the runtime changes a private local copy. The collaboration MCP coordinates deterministic Graph download, validation, and same-item publication; it is not a second Office editor in the gateway.

04 / Document edit and publication lifecycle



A changed source is re-read. Only reconstructible guarded edits can rebase; opaque edits fail closed.

Preparation and model reasoning happen once. A background publication retry executes a storage workflow, not a repeated “try again” prompt.

Successful same-item publication preserves the document's identity and collaboration context. A copy is explicitly identified as a different item with different sharing, comments, and history. If the original remains locked, the current implementation offers a bounded retry window and then a verified shared copy; it never reports the original as updated when the upload failed.